

It's starting off carrier wars all over the country—but can WLL-CDMA services possibly be the ones to get India connected?

It's not what kind of network you use, it's about what the network can offer you. We've all heard about how the last mile connectivity can turn out to be hideously expensive. Cell phones are continuing on their onslaught to takeover the world, and though it has been several years since the introduction of cellular communication services in India, it still hasn't brought the vast majority of us Indians closer to a phone. We still have a very low tele-density. Here comes the new kid on the block, WLL, (standing for Wireless in Local Loop), it may be just be our networking solution. It's been touted as the one that will get one billion Indians connected to a phone line. The introduction of WLL services has also highlighted a newer digital communications technology, the CDMA (Code Division Multiple Access) wireless technology.

ing to dig and layout new cabling.

Since WLL is a wireless access method that uses radio waves to connect the subscriber to a local central office, it can be implemented alongside a regular PSTN (Public Switched Telephone Network), or be combined with digital wireless technologies, such as CDMA (Code Division Multiple Access) and GSM. The use of a WLL system alongside a CDMA network has been very successful worldwide and for a good reason. Here in India, MTNL started WLL services in December 2001 and BSNL has been toying with it for quite some time now. But, what has really made WLL a buzzword is the entry of telecom biggies, such as Reliance Infocomm and the Tata group. WLL-based networks are coming up nationwide and promise quick and immediate access to telephone services and cheaper calling rates. They also offer

Where there is a **WLL** there is a **way**...

WLL it, won't it?

WLL is a new communications access method that uses radio waves for the transmission of information between customers and service provider sites ('the exchange' or central office). This is unlike the traditional methods where a fixed copper or optical fibre line is used to connect the subscriber to the nearest central office. In the traditional copper-wire setup, the two-wire connections between each subscriber's telephone and the local central office, is known as the local loop. With the WLL link, the service provider can avoid the expense of laying out copper cables to the subscriber's home. While most urban areas are well laid out with the requisite cabling, it is not very hard to implement WLL services in areas that are inaccessible or remote. This is where the beauty lies. WLL could step in to provide connectivity in rural areas, as WLL networks are easier to set up than regular landline networks. With a WLL network in place, the service providers can increase the capacity of the network without hav-

the likes of broadband and multimedia-rich content on your handheld, when the upgrade to the next generation standards takes place. With the WLL link in place, subscribers might soon have the best of both worlds—a phone on demand and access to a slew of multimedia-rich content, making the phone an entertainment device. As most of the WLL services in India are being launched on the CDMA platform, we get the advantage of using the CDMA technology—a comparatively newer technology than the existing GSM standard.

In the mobile, wire-free world of today, the dominant technologies are GSM (Global System for Mobile communications) that started out as an European standard, and is now commonly used in Europe, Asia and Africa. The other is CDMA, which has been implemented in North America, Japan and Korea. Wireless communication systems are designed to use the radio spectrum to send and receive information through the air. While the spectrum may appear to be a